

Chapter 30 — Self-Description and Constitutional Mirror

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.2
Scope	Maintain a verified model of HAL's identity, authority, capabilities, limitations, Presences, topology, health, Treaties, and operating state.
Constitutional basis	Decisions 7, 28, 38, 40, 45, 47, 49, 51
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Maintain a verified model of HAL's identity, authority, capabilities, limitations, Presences, topology, health, Treaties, and operating state.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Self Model Service	Canonical self-description assembled from governed registries and evidence
Consistency Checker	Declared-versus-observed identity, authority, policy, capability, and topology validation
Constitutional Mirror Service	Signed human-readable snapshots and continuity comparison
Self Explanation Service	Audience-appropriate answers grounded in verified self-state

3. Core State and Records

Object	Required content
Self Model	Identity, Owner, Constitution, authority, capabilities, limitations, Presences, nodes, health, policies, and Treaties.
Limitation Record	Unavailable/unauthorized capability, cause, confidence, duration, alternatives, and evidence.
Identity Drift Incident	Expected state, observed divergence, evidence, containment, and resolution.
Constitutional Mirror	Signed versioned snapshot of Constitution, identity, authority, trust, intent, knowledge, topology, health, and pending work.

4. Runtime Workflow

1. Read identity, authority, policy, capability, Presence, cluster, Treaty, health, and recovery registries.
2. Assemble a versioned Self Model without asking an LLM to invent missing self-knowledge.
3. Compare declared state with independently observed runtime and durable state.
4. Raise and contain identity drift, unauthorized capability, or unexplained policy divergence.
5. Answer self-questions from verified records and state explicit limitations or unknowns.
6. Generate signed Constitutional Mirrors on schedule and before/after material change or recovery.
7. Compare Mirrors to establish explainable continuity across implementation replacement.

5. Interfaces and Contracts

Commands, queries, events, records, and evidence are typed, versioned, authenticated, authorized, correlated, and classified. Mutation is accepted only by the authoritative owner. Every external or reality-changing boundary carries identity, policy, authority, provenance, integrity, time, and transaction context.

6. Failure and Recovery

- Registry disagreement: report the conflict and lower self-description confidence.
- Mirror generation fails: block any workflow that requires a verified continuity checkpoint.
- Self Model stale: display last verified time and do not claim current capability.
- Identity drift: enter constitutional investigation rather than normal self-healing.

7. Constitutional Guarantees

- HAL describes itself from verified evidence, never assumption or prompt identity.

- Implementation may change while constitutional identity and governed continuity remain intact.
- Limitations are first-class self-knowledge and never hidden for conversational convenience.
- A Constitutional Mirror explains a backup or recovery point; it does not replace it.

8. Prohibited Behaviors

- No component may silently broaden or narrow constitutional authority, reinterpret Owner intent, erase dissent, or promote derived state to authority.
- No failure, degraded dependency, emergency, or optimization may silently change identity, policy, trust, authentication, evidence, or verification rules.
- No model, service, provider, node, Presence, credential, relationship, environment, or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records identity, correlation, authority/policy context, inputs, state versions, decisions, limitations, failures, recovery, and outcomes. Verification is risk-scaled, reproducible, and stored as durable evidence. Status and explanation derive from governed state rather than model assertion.

10. Source Alignment and Review

This chapter implements Decisions 7, 28, 38, 40, 45, 47, 49, 51. Final source-alignment and constitutional disposition is recorded in the batch audit reports; embedded approval text is not treated as review evidence.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- Embedded author approval is not conformance evidence. This chapter's constitutional status is determined only by the independent whole-book audit.
- No substantive constitutional or cross-chapter correction was required in this edition.